Interpretable Respiratory Sound Analysis

Project Background

This note explains what the project is, where the recordings come from, and why your listening judgment is genuinely useful to it.

What the project is

Most machine learning models for detecting disease from lung sounds work as a black box: audio goes in, a diagnosis comes out, and there is no way for a clinician to see or correct the reasoning in between. This project builds a diagnosis pipeline that instead routes every decision through a small set of **named acoustic concepts** — crackle, wheeze, and related adventitious sounds — computed directly from the audio signal using established signal-processing methods, not learned from opaque features. The disease prediction is only allowed to depend on these concept values, so a clinician can, in principle, inspect which sounds the model believed it heard, and correct one if it is wrong.

The immediate technical question is simpler than that final goal: **do the automated concept detectors actually measure what they claim to measure?** A detector labeled "crackle" is only useful if it responds to crackles specifically, not to breath sounds in general. That is what this listening task is designed to check.

The dataset

The audio comes from the **ICBHI 2017 Respiratory Sound Database**, a public research corpus of 920 recordings from 126 patients, collected with digital stethoscopes across multiple clinical sites. Each recording is segmented into individual respiratory cycles, and each cycle is already labeled by the dataset's original creators for the *presence* of crackles and wheezes — but not for finer distinctions such as fine versus coarse crackle, or wheeze pitch. All recordings are anonymized archival research data; there is no patient-identifying information, clinical history, or imaging attached to any clip.

Why we need your review

The dataset's own labels only go so far: they say whether a crackle is present, not what kind. Our automated detectors compute several finer-grained acoustic properties, but without an independent human judgment there is no way to know whether those finer measurements are trustworthy. Your listening task provides exactly that independent check — a blind, expert judgment on a sample of clips, made without seeing what our detectors concluded, so that your answers can be compared fairly against both the automated output and the dataset's own labels. This also lets us tell apart two different explanations for any disagreement we see: that our detectors need work, or that the dataset's original labels are themselves imperfect — both are useful things to know, and both are publishable findings in their own right.

Publication scope

This work is being carried out as a university machine learning research project, with the explicit goal of submitting the results to a peer-reviewed biomedical engineering journal. Your listening judgments, if used, would be described in the resulting paper as an expert validation step, and you would be acknowledged by name (with your permission) as the clinician who performed it. No patient data, institutional affiliation, or clinical opinion beyond your acoustic judgments on these anonymized clips would be attributed to you.

Thank you again for lending your expertise to this — it directly strengthens the scientific credibility of the project.